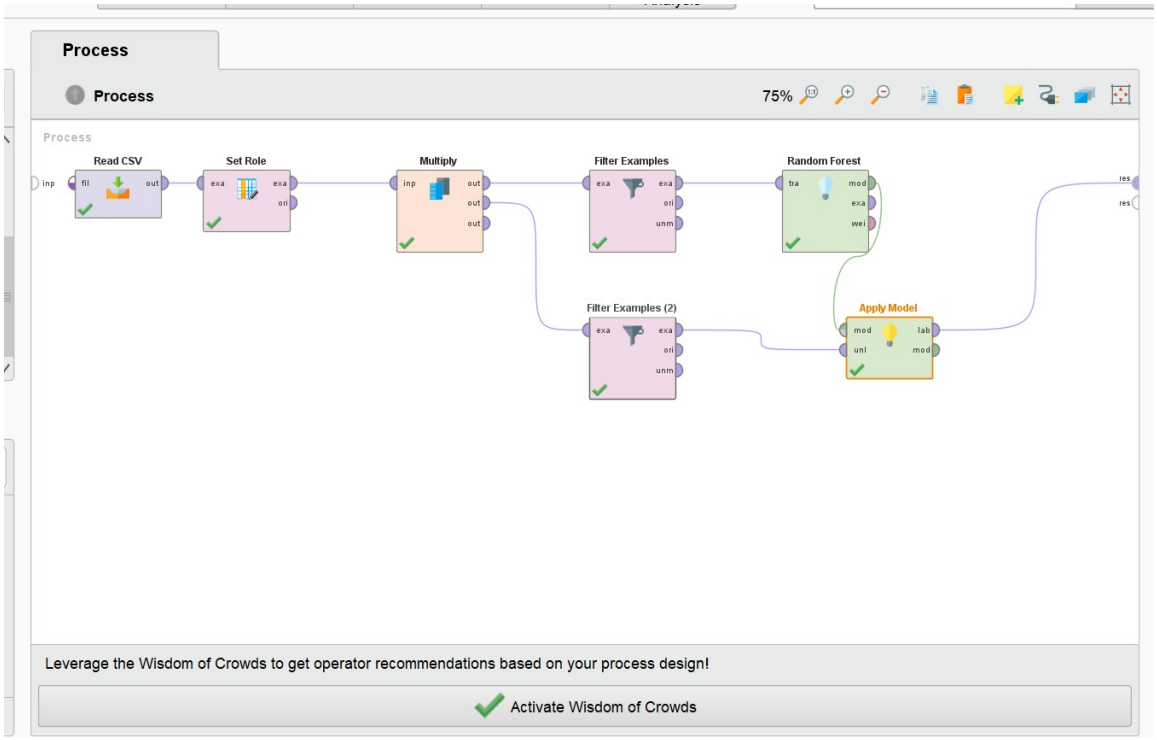
Shape Prediction using Random Forest in Altair AI Studio

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Task

The data has about 15 rows with the columns **shape**, **color**, **outline** and **dot**, where the shape is known. Two more rows have color, outline and dot but their **shape is missing (?)**. The task is to learn from the 15 known rows and **predict the shape** of the 2 unknown rows.

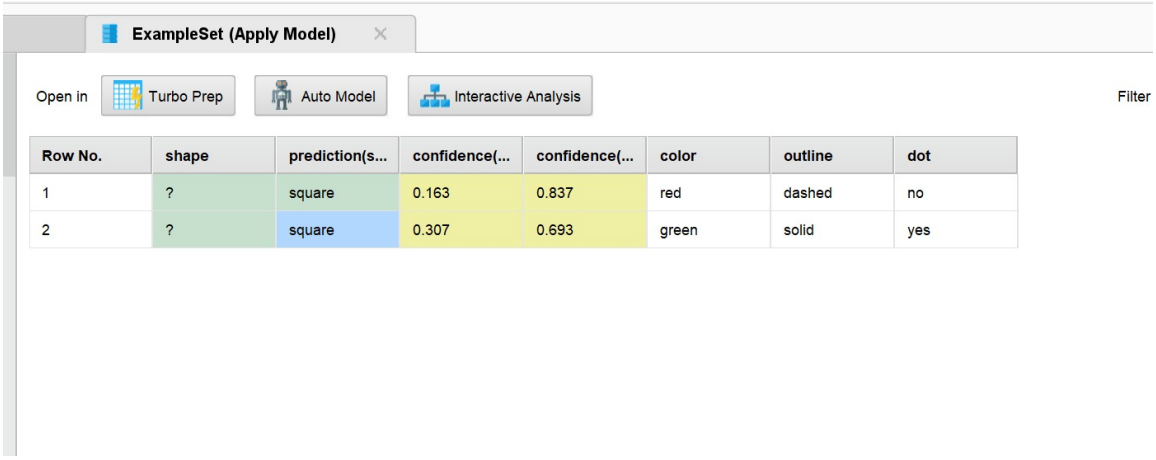
Process Design



Process design in Altair AI Studio.

1. Read CSV: loads the data file.
2. Set Role: sets **shape** as the label, the column to predict. Color, outline and dot stay as regular inputs.
3. Multiply: makes two copies of the data, so one copy can be used for learning and the other for prediction.
4. Filter Examples (top): keeps only the rows where the shape is known (the 15 rows) as training data.
5. Random Forest: learns from these rows by building many decision trees and combining their results.
6. Filter Examples (2): keeps only the 2 rows where the shape is missing.
7. Apply Model: takes the trained Random Forest model (mod) and applies it to those 2 rows (unl) to predict their shape.

Result



Row No.	shape	prediction(s...	confidence(...	confidence(...	color	outline	dot
1	?	square	0.163	0.837	red	dashed	no
2	?	square	0.307	0.693	green	solid	yes

Apply Model output: predicted shape for the 2 unknown rows.

The shape column still shows **?** because it was unknown. The new **prediction** column gives the answer, and the **confidence** columns show how sure the model is for each shape.

• Row 1 (red, dashed, no dot): predicted **square** with confidence 0.837.
• Row 2 (green, solid, dot): predicted **square** with confidence 0.693.

Conclusion: using a Random Forest trained on the 15 known rows, both unknown shapes were predicted to be **square**.